

UCCN 2003

UCCN 2243

TCP/IP Internetworking, Internetwork Principles & Practices (Lecture 03)

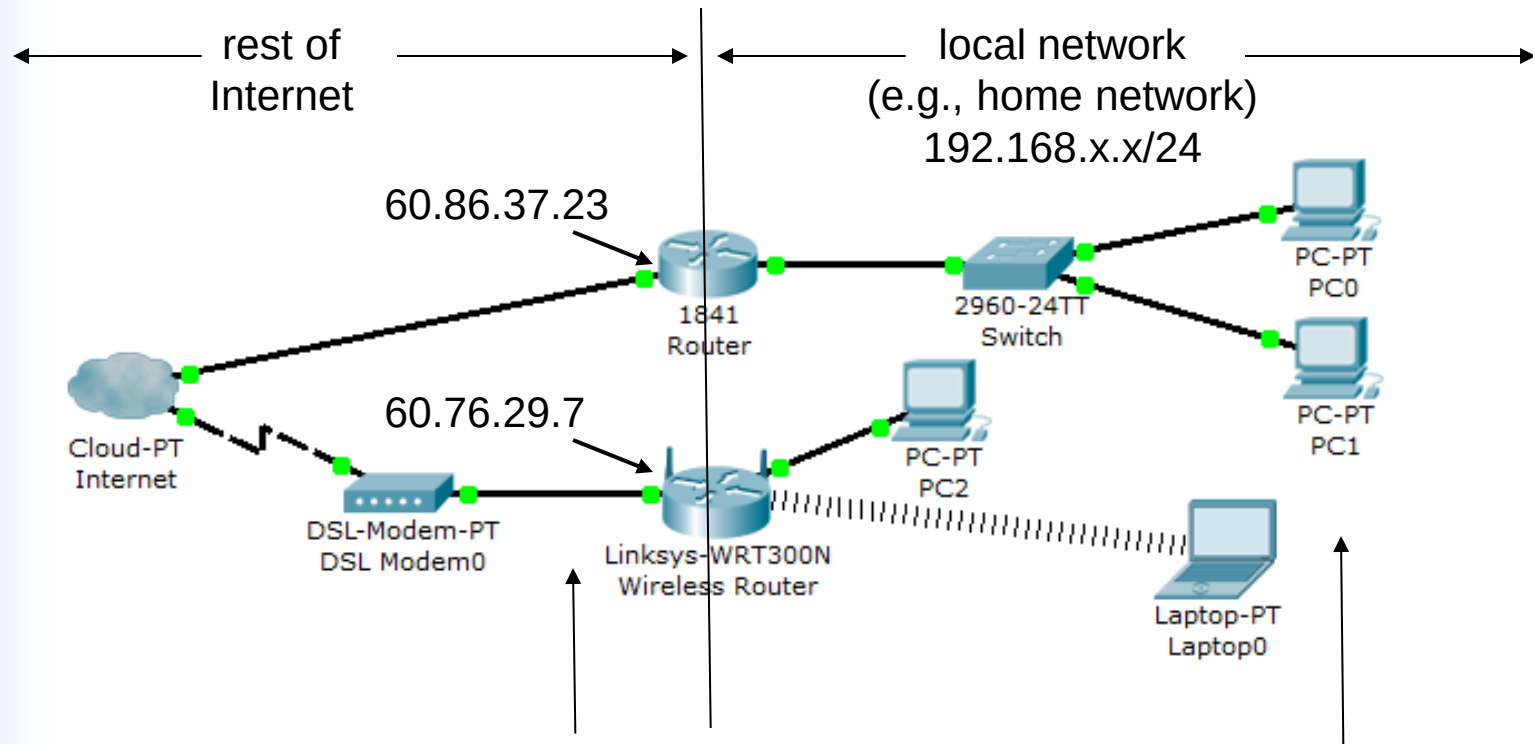
Network Address Translations (NAT)

Introduction & Overview of NAT

What is NAT?

- Network address translation (NAT) is
 - the process of modifying the network address in IP packet headers while in transit across a routing device.
- NAT was invented in 1994.
- NAT is used in conjunction with network masquerading or IP masquerading
 - which is a technique that hides an entire IP address range (usually private network IP addresses), behind another single IP address (often public address).
 - a web browser in the masqueraded network can browse a website outside, but a web browser outside could not browse a web site in the masqueraded network.

Example of NAT Mechanism



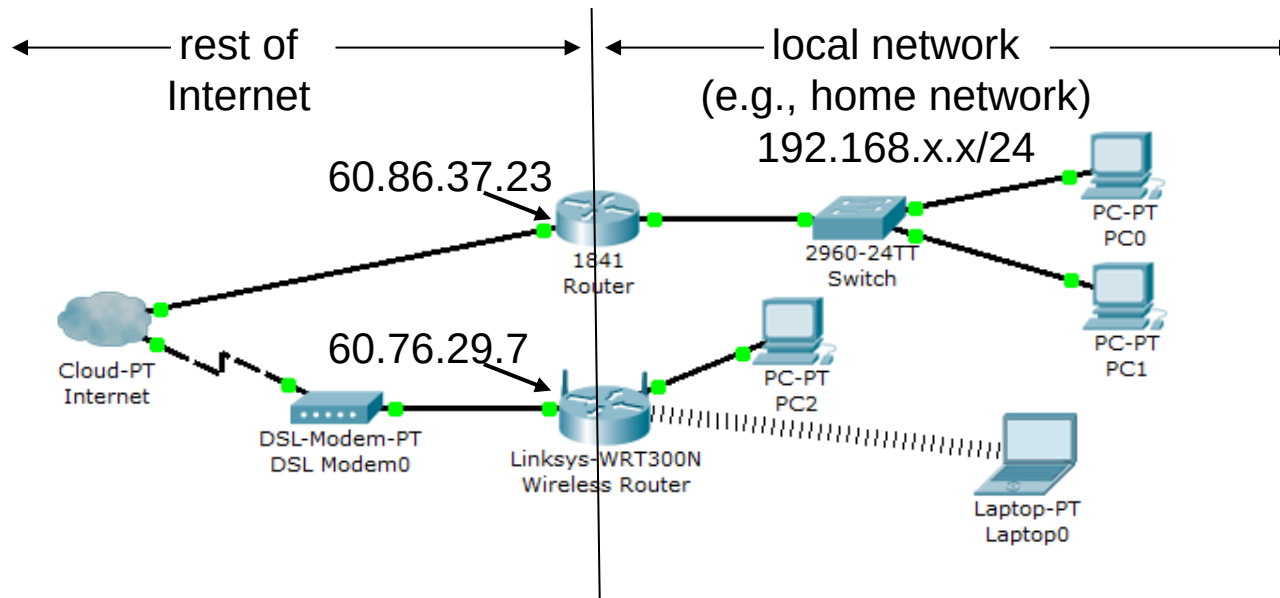
All datagrams *leaving* local network have same single IP address (by NAT process):
Wireless Router => 60.76.29.7 for PC2, Laptop0
Router => 60.86.37.23 for PC0, PC1

Datagrams with source or destination in this network have 192.168.x.x/24 address for source

Why do we need this “NAT”?

- NAT is used for three major reasons:
 - IPv4 address exhaustion
 - Masquerading for security purpose
 - TCP load sharing
- NAT for alleviating the consequences of IPv4 address exhaustion.
 - It has become a standard, indispensable feature in routers for home and small-office Internet connections.
 - One public IP can be used by thousands of private network computers.
- NAT as IP masquerading
 - Obscures an internal network's structure,
 - All network traffic appears to outside network as if it is originated from the one IP address of a router.
- NAT for TCP load sharing
 - Useful for server farm
 - A few servers with similar functions represented by one single IP address.

NAT - Important Skill for IPv4

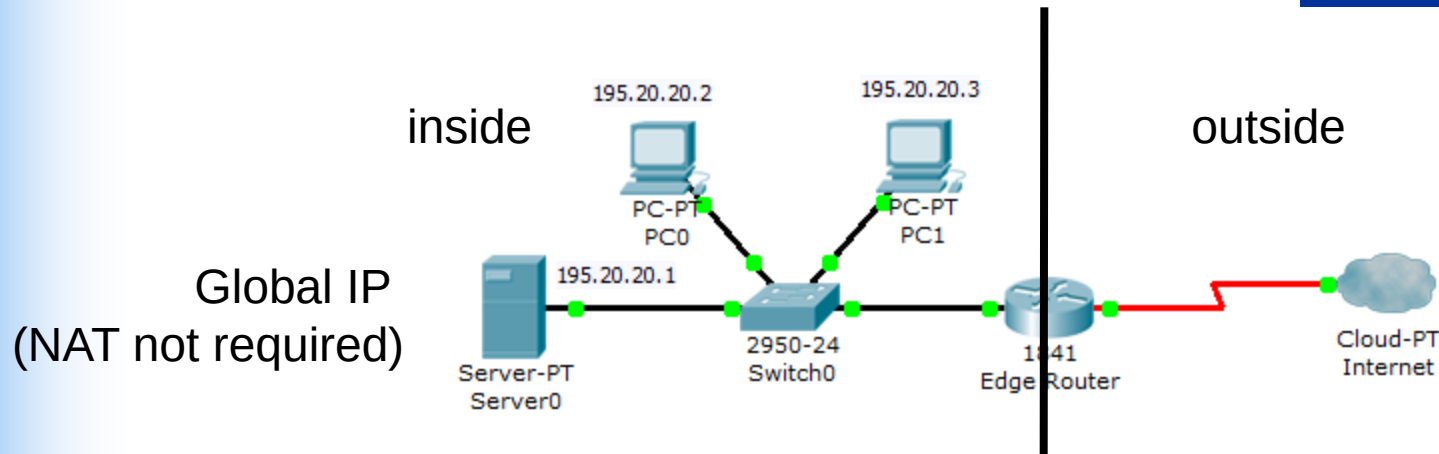


- NAT is a skill you **MUST** have in order to:
 - configure the router next to the Internet, which the router contain the public IPv4 address.
 - normally, we named this router as “edge router”.
 - IPv4 Internet currently **CAN'T** function without NAT
 - IPv4 simply **DOES NOT HAVE ENOUGH** addresses to cater for every end-devices in the world.
 - For example, a lot of institutions has only 1 public IP to cater for thousands of PCs behind the edge router.

Issue of IPv4 Address Exhaustion

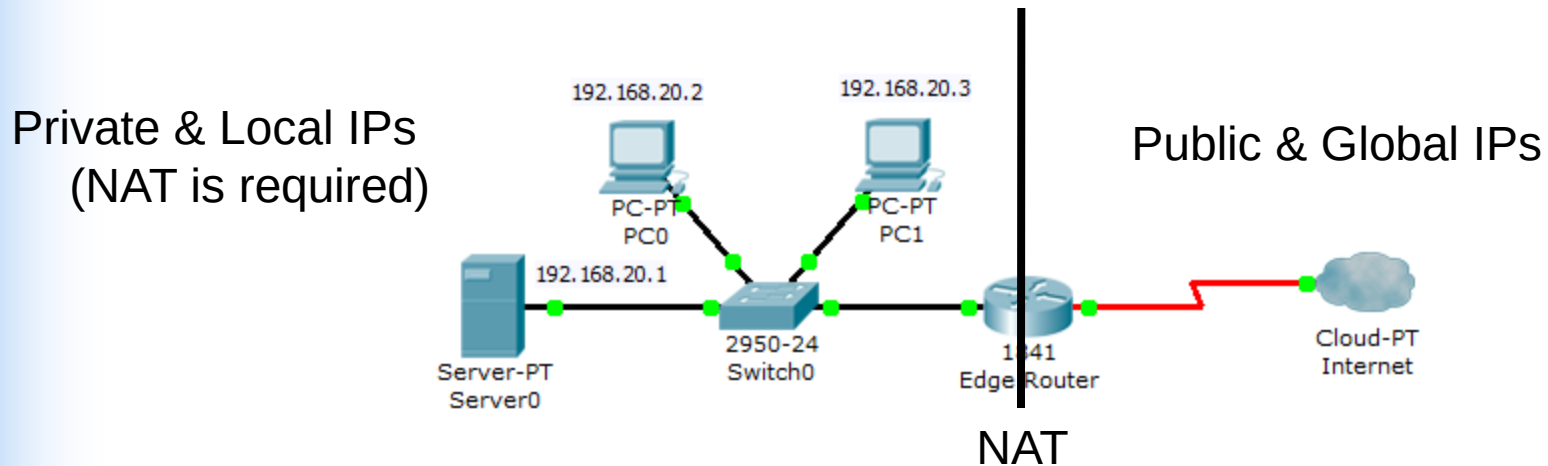
- IPv4 address exhaustion
 - is the decreasing supply of unallocated IPv4 addresses available at the Internet Assigned Numbers Authority (IANA).
 - NAT is so important that the current Internet based on IPv4 can't function without it due to address exhaustion.
- Some methods of mitigation of IPv4 address exhaustion have been
 - Classless Inter-Domain Routing (CIDR)
 - Network address translation
 - Use of private network addressing
 - Virtualization
 - Name-based virtual hosting of web sites
 - Multiple host names using the same IP address

Terms of NAT – Inside & Outside



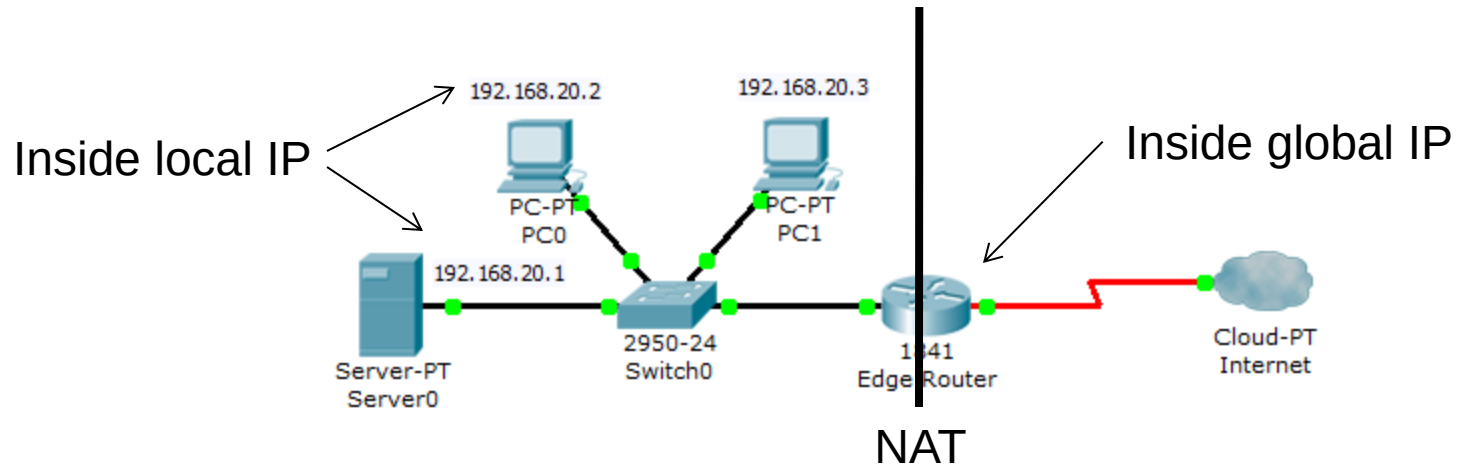
- There are four important terms in NAT:
 - First two terms are inside and outside
- Inside is our own network or LAN.
- Outside is the rest of the world, the Internet.
- If you do not use NAT, we need to use public IP for all the end devices.

Terms of NAT – Local & Global



- There are four important terms in NAT:
 - Second two terms are local and global
- Local is our own network or LAN configured with private IP addresses.
- Global is the rest of the world, the Internet using public IP addresses (assigned through IANA).

Terms of NAT - Inside Local & Inside Global



- Using a NAT enable router, we are using inside local addresses which are private IP address and are NOT unique in the world.
- The inside local address must be translated to inside global address via NAT process of the router.
- Inside global addresses are public IP address seen from the “outside”.
 - But these addresses “belongs” to our hosts, that’s why they are called “inside” global.

Terms of NAT - Summary

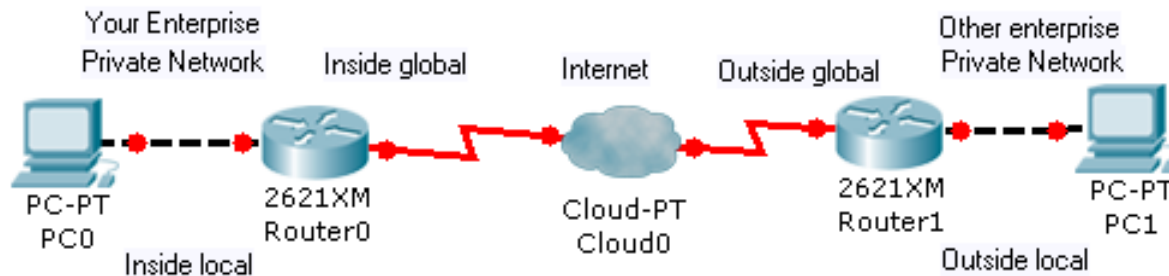
This NAT table is maintained inside the router.

Inside local IP address		Inside global IP address
10.1.1.1	↔	193.99.99.1
10.1.1.2	↔	193.99.99.2
10.1.1.3	↔	193.99.99.3
10.1.1.4	↔	193.99.99.4

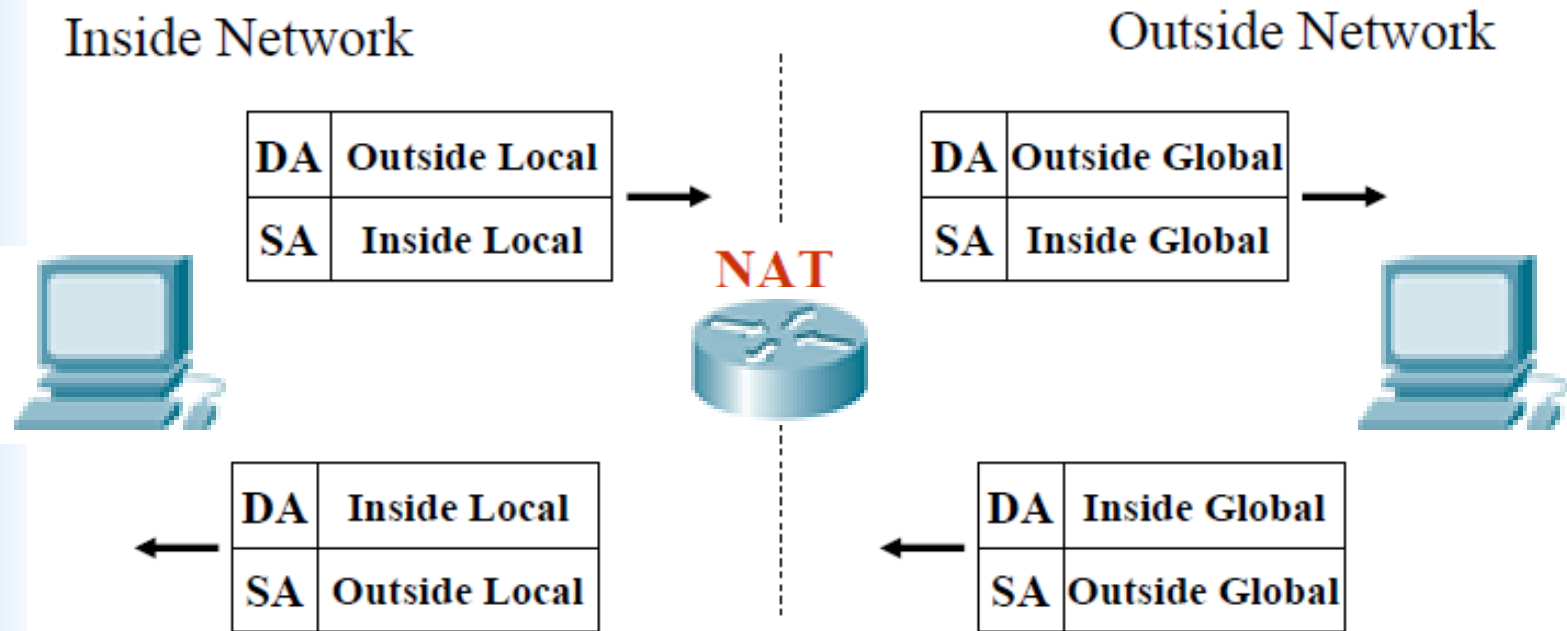
- **Local versus global**
 - NAT router is responsible to translate global address to local OR local addresses to global.
- **Inside versus outside**
 - Inside means your network
 - Outside means network beyond the NAT router.

Terms of NAT - Summary

- Inside local
 - IP address (normally private IP) assigned to a host in “your” private enterprise network. Or “inside private”
- Inside global
 - The public global IP address(es) of your enterprise
- Outside global
 - The public global IP address(es) of other enterprise’s network.
- Outside local
 - For this level of study, outside local = outside global



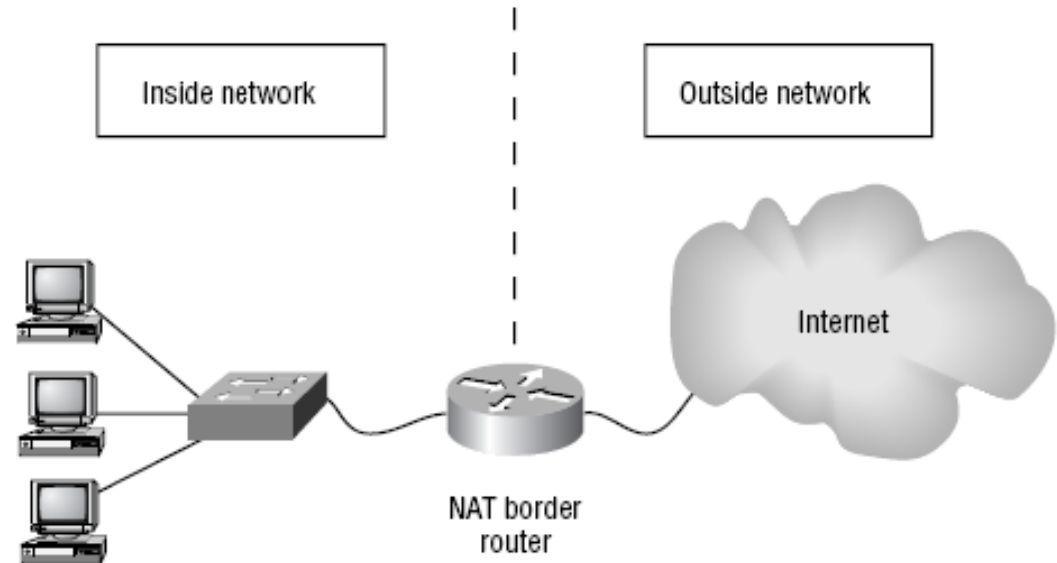
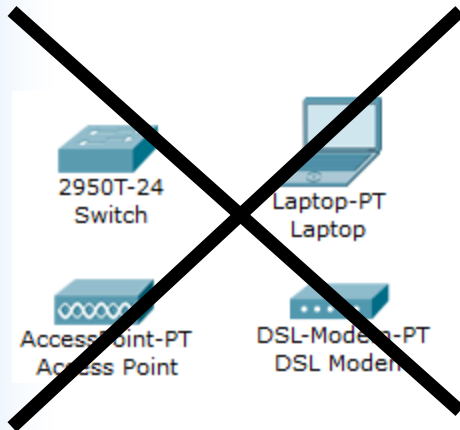
Terms of NAT in action



- DA = Destination IP address
- SA = Source IP address

NAT is done by Router

Do not perform NAT



- NAT is normally performed at the border router to the Internet.
 - Any layer-3 device: router & layer-3 switch have the ability to perform NAT
- Layer-2 Switch, wireless access point, modems and end devices DO NOT perform NAT.

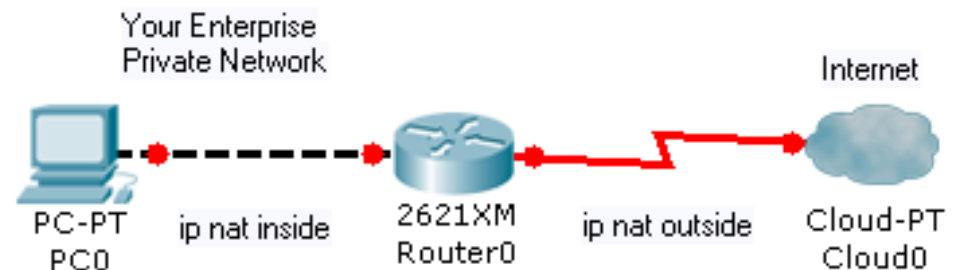
Four Types of NAT

- Static Network Address Translation (static NAT)
 - 1 private IP to 1 global IP address translation
- Dynamic Network Address Translation (dynamic NAT)
 - Many private IP to many global IP address translation (a pool of IP)
- Port Address Translation (PAT)
 - Many private IP to 1 global IP address translation.
 - Is also called NAT overloading.
- Port Forwarding
 - Accessing a inside local network service from outside global host.

Initial Configuration of NAT

First Configuration of NAT

- When you configure NAT on a Cisco router,
- You need to first tell the router
 - The private network side (inside)
 - The public network side (outside)
- The inside/outside is defined at the interface.
- You have to do this step first before setting static NAT, dynamic NAT, PAT, and port forward

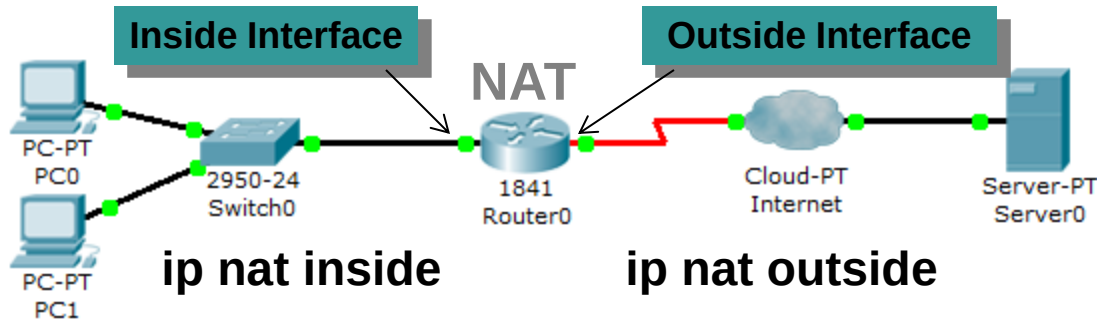


```
Border(config)#interface ethernet0
Border(config-if)#ip nat inside
Border(config-if)#exit
Border(config)#interface serial0
Border(config-if)#ip nat outside
Border(config-if)#exit
Border(config)#
```

Inside/Outside interface

Inside Network

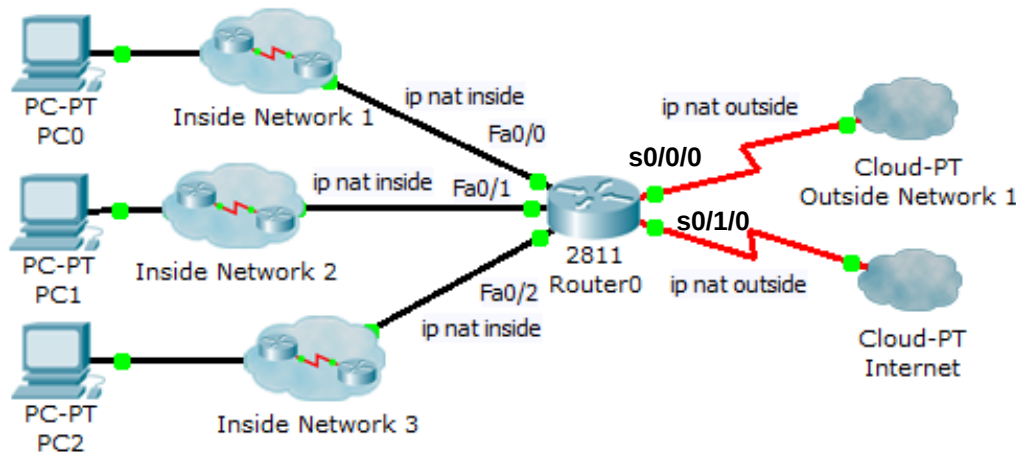
Outside Network



```
Router(config-if)#ip nat inside  
Router(config-if)#ip nat outside
```

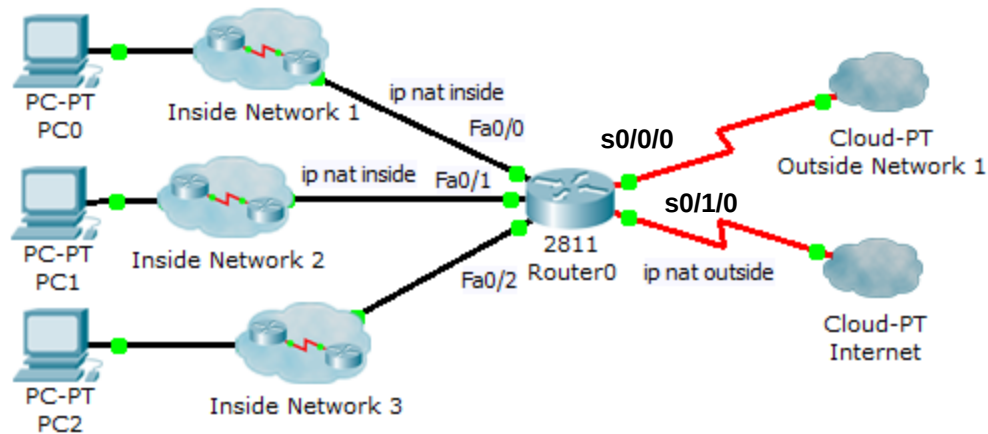
- Any interface on the router (e.g. serial, fast Ethernet, Ethernet) can be defined as inside or outside
 - An interface **CAN'T** be both inside and outside at the same time.
- Translations occur only from inside to outside interfaces or vice versa—never between the same type of interface (e.g. inside to inside, outside to outside).

Multiple Inside Multiple Outside



- Any one interface of the router can't have both inside and outside.
- But one router can be configured to have multiple “ip nat inside” and multiple “ip nat outside”.

NAT pair : inside to outside



- NAT can only occur with a “pair” of “ip nat inside” and “ip nat outside”.
 - Please note that you need to set the static NAT or dynamic NAT or PAT.
- Hence, in the above network, NAT can only happen in packets traversing:
 - From Fa0/0 to s0/1/0
 - From Fa0/1 to s0/1/0
- No NAT will occur:
 - From Fa0/0 to s0/0/0
 - From Fa0/1 to s0/0/0
 - From Fa0/2 to s0/1/0

Inside/Outside Status

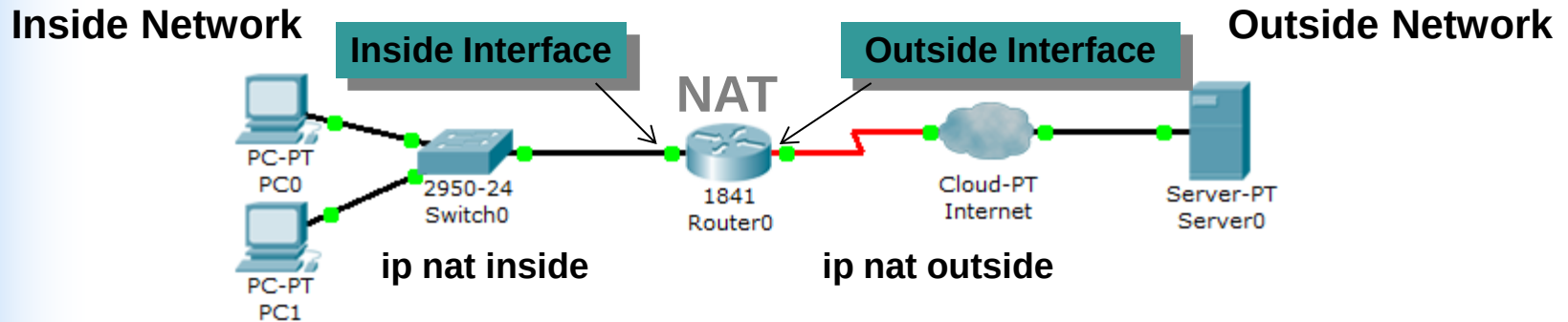
```
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#ip nat outside
```

```
Router(config-if)#int fa0/1
Router(config-if)#ip nat inside
Router(config-if)#ip nat outside
Router(config-if)#ip nat inside
```

```
!
interface FastEthernet0/0
  no ip address
  ip nat outside
  duplex auto
  speed auto
!
interface FastEthernet0/1
  no ip address
  ip nat inside
  duplex auto
  speed auto
!
```

- Since each interface only can be either “ip nat inside” or “ip nat outside”
- The “inside” or “outside” status can be overwritten based on the “last inside/outside” command.
- Use “show run” command to check the status of “ip nat inside/outside” contained in the interfaces.

NAT - Steps of “to and fro”



1. Inside host has to start the connection first

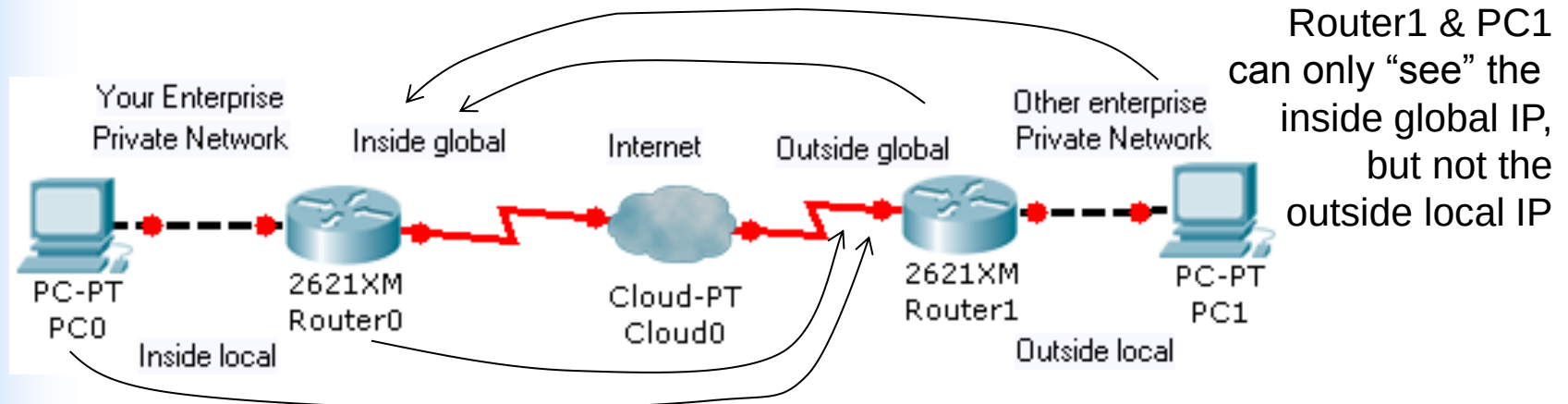
2. Inside source IP changed to public IP
3. A translation entry is recorded at router

5. Based on the NAT entry, the router convert the destination IP back to inside host IP

4. Outside host sends back to public destination IP

- In NAT, inside host always has to initiate the connection to the outside host first.
- The NAT will record a “translation” entry.
- So that, the packet of outside host knows “how to come back”.
- An outside host does not “see” the hosts in a inside network.
- The outside host only see the public IP after the NAT.

Private Network to Private Network



Router0 & PC0 can only “see” the outside global IP, but not the outside local IP.

- Hosts in a private network (especially with private network IP, e.g. 192.168.x.x) can't directly connect to another host within another private network via the Internet, though routers at both sides have NAT.
- In the above diagram
 - PC1 can only “see” the global IP of Router0, but not the local IP of PC0
 - PC0 can only “see” the global IP of Router1, but not the local IP of PC1

#Quick Quiz#

- Which IP does the NAT normally change?
 - Source IP or
 - Destination IP?

Answer

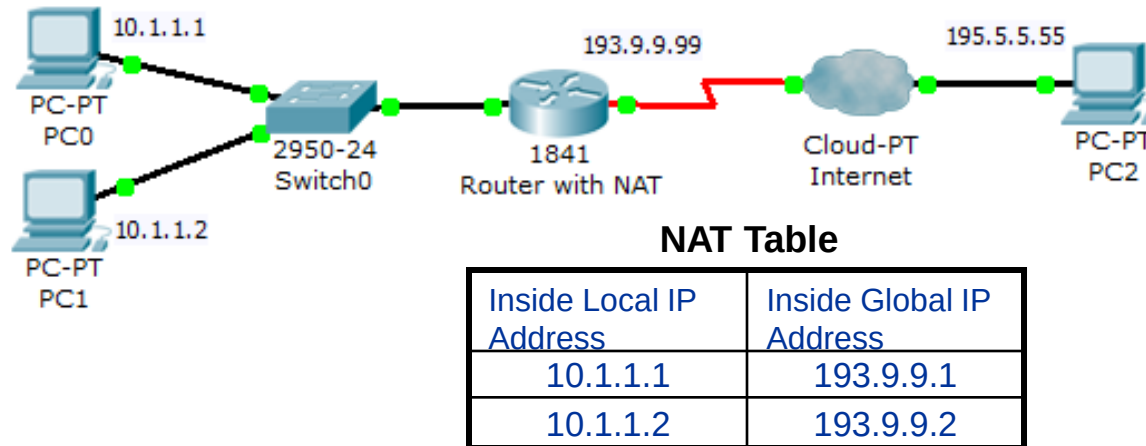
- Both ways.
 - Depending on your direction.

After interfaces....

- Defining the inside interfaces and outside interfaces are just the first steps of NAT.
- Next, we need to determine to use one of the following NAT methods to perform the translations.
 - Static Network Address Translation (static NAT)
 - Dynamic Network Address Translation (dynamic NAT)
 - Port Address Translation (PAT)
 - Port Forwarding

Static NAT

Working Principle of Static NAT



- Static NAT is designed to allow one-to-one mapping of local and global addresses.
- Router has been configured with static NAT:
 - User at PC0 10.1.1.1 opens a connection to PC2 195.5.5.55
 - Packet received by Router and Router check its NAT table.
 - Router replaces the source address with the inside global IP found in the NAT table.

Configuring Static NAT

```
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#int s0/1/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source static 10.1.1.1 193.9.9.1
Router(config)#ip nat inside source static 10.1.1.2 193.9.9.2
```

- Use “ip nat inside source static *inside_local_ip* *inside_global_ip*” to configure static NAT.
- Use “show ip nat translation” command to check the NAT table.

```
Router(config)#exit
Router#show ip nat translation
```

Pro	Inside global	Inside local	Outside local	Outside global
---	193.9.9.1	10.1.1.1	---	---
---	193.9.9.2	10.1.1.2	---	---

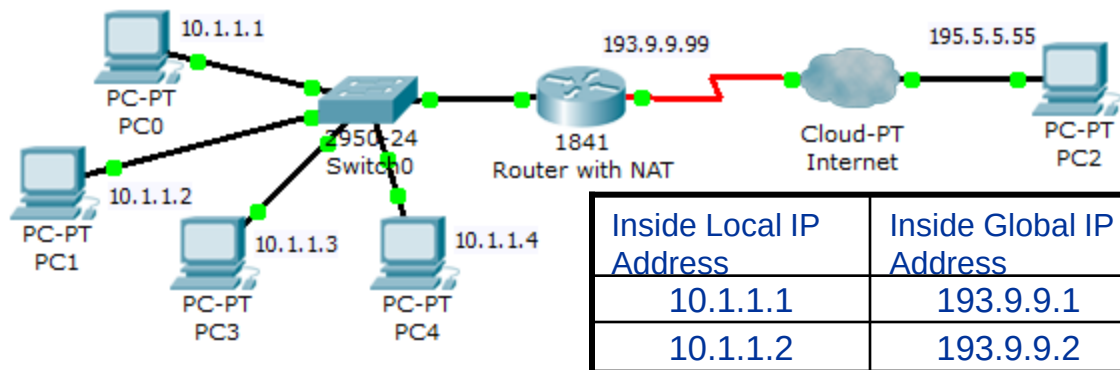
Comments on Static NAT

inside network source IP Static NAT original IP translated IP

```
Border(config)#ip nat inside source static 10.1.2.25 200.1.1.25
Border(config)#
```

inside local inside global

- In static NAT, if you have 10 private IP, you need 10 global IP in static NAT.
- NAT process will always translate the same “pair” of IP.
- In the following network, no NAT for PC3 and PC4, since the NAT table does not contain their IP address.



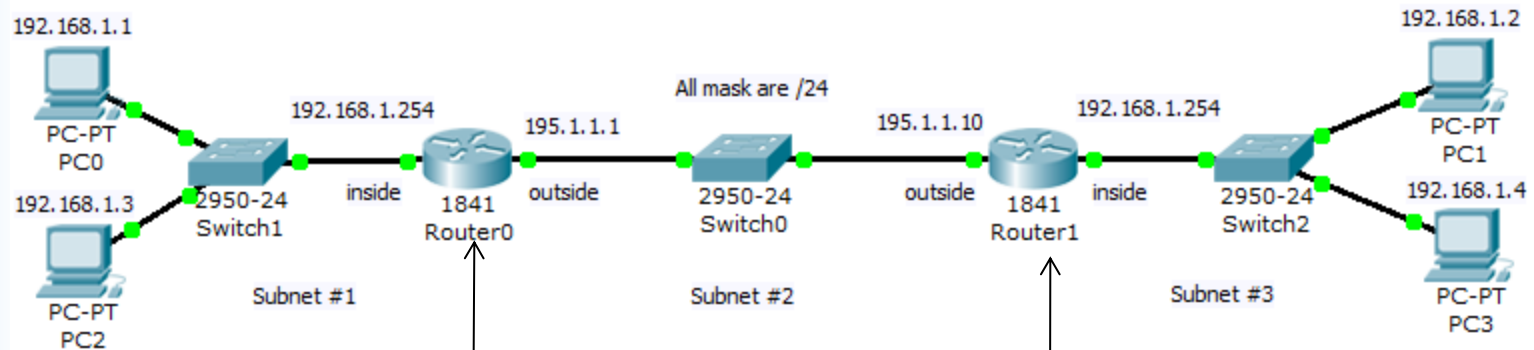
NAT Table

#Quick Quiz#

```
!  
ip nat inside source static 192.168.1.1 200.1.1.173  
ip nat inside source static 192.168.1.2 200.1.1.174  
ip nat inside source static 192.168.1.3 200.1.1.175  
ip nat inside source static 192.168.1.4 200.1.1.176  
!
```

- What IP does the NAT change to, given the following IP:
 - 192.168.1.3

Quick Quiz



```
ip nat inside source static 192.168.1.1 195.1.1.5  
ip nat inside source static 192.168.1.3 195.1.1.7
```

```
ip nat inside source static 192.168.1.2 195.1.1.14  
ip nat inside source static 192.168.1.4 195.1.1.16
```

- PC3 tries to ping 195.1.1.5
- Write the source IP and destination IP of this ping packet as it travels through three subnets and back.

Answer

- SIP = source IP; DIP = destination IP
- To:
 - Subnet #3:
 - SIP = 192.168.1.4; DIP = 195.1.1.5
 - Subnet #2:
 - SIP = 195.1.1.16; DIP = 195.1.1.5
 - Subnet #1:
 - SIP = 195.1.1.16; DIP = 192.168.1.1
- Back:
 - Subnet #1:
 - SIP = 192.168.1.1; DIP = 195.1.1.16
 - Subnet #2:
 - SIP = 195.1.1.5; DIP = 195.1.1.16
 - Subnet #3:
 - SIP = 195.1.1.5; DIP = 192.168.1.4

Answer

- NAT table of Router0 and Router1, after PC3 ping 195.1.1.5

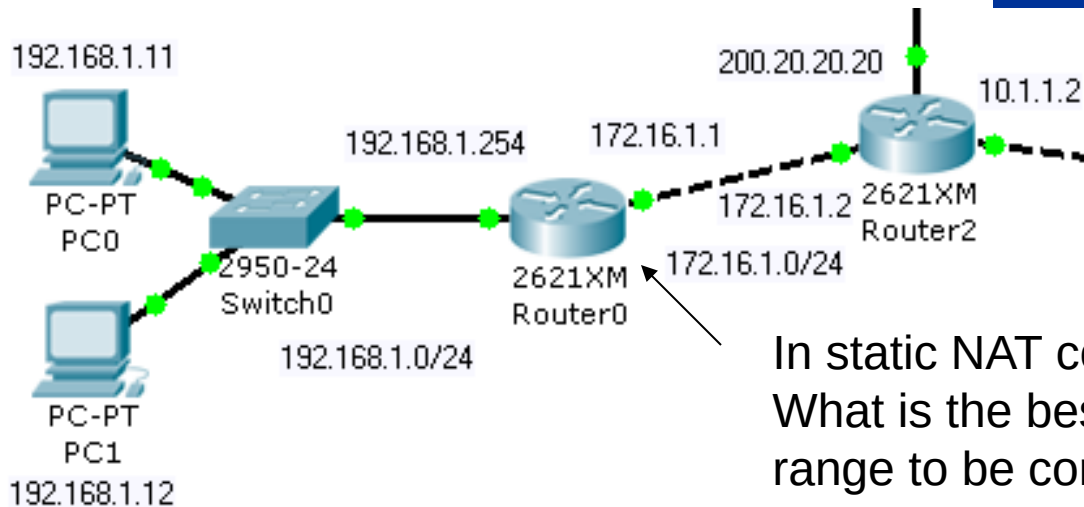
```
Router0#show ip nat translation
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	195.1.1.5:11	192.168.1.1:11	195.1.1.16:11	195.1.1.16:11
icmp	195.1.1.5:12	192.168.1.1:12	195.1.1.16:12	195.1.1.16:12
icmp	195.1.1.5:13	192.168.1.1:13	195.1.1.16:13	195.1.1.16:13
icmp	195.1.1.5:14	192.168.1.1:14	195.1.1.16:14	195.1.1.16:14
---	195.1.1.5	192.168.1.1	---	---
---	195.1.1.7	192.168.1.3	---	---

```
Router1#show ip nat translation
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	195.1.1.16:11	192.168.1.4:11	195.1.1.5:11	195.1.1.5:11
icmp	195.1.1.16:12	192.168.1.4:12	195.1.1.5:12	195.1.1.5:12
icmp	195.1.1.16:13	192.168.1.4:13	195.1.1.5:13	195.1.1.5:13
icmp	195.1.1.16:14	192.168.1.4:14	195.1.1.5:14	195.1.1.5:14
---	195.1.1.14	192.168.1.2	---	---
---	195.1.1.16	192.168.1.4	---	---

Static NAT Design Issues



- Given the above scenario, in which 192.168.1.0/24 needs to undergo a static NAT in Router0, what should be the IP address range to be converted?
- Answer: Recommended IP range will be subnet IPs where the “outside interface” belongs to, but do not include the IP address of the “outside interface”.
- In the above example, the recommended IP range is: 172.16.1.3 to 172.16.1.254
- If you use IP range other than 172.16.1.3 to 172.16.1.254 in the above example, the outgoing packet may not be able to come back due to routing issues.

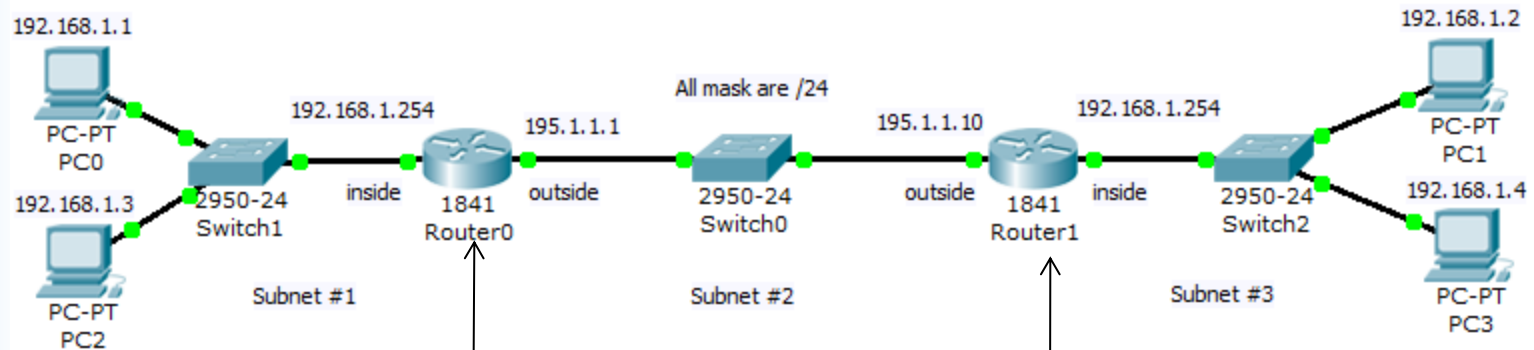
Comments on ARP Table

```
Router#show arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	10.1.1.2	-	00D0.5852.8A01	ARPA	FastEthernet1/0
Internet	172.16.1.1	3	0004.9AAE.D701	ARPA	FastEthernet0/1
Internet	172.16.1.2	-	0000.0CAA.B102	ARPA	FastEthernet0/1
Internet	172.16.1.11	1	0004.9AAE.D701	ARPA	FastEthernet0/1
Internet	172.16.1.12	0	0004.9AAE.D701	ARPA	FastEthernet0/1
Internet	200.20.20.1	0	0090.2BB1.D34B	ARPA	FastEthernet0/0
Internet	200.20.20.2	2	0050.0F03.DB75	ARPA	FastEthernet0/0
Internet	200.20.20.20	-	0000.0CAA.B101	ARPA	FastEthernet0/0

- Continue from previous slide....
- Assume that the PCs has initiated the communication:
 - If we check the arp table, we find that the arp table contains entries with the same MAC address, but different IP.
 - This explains why the router knows how to send back to the same “out interface” of the router.
 - The arp table entry was formed with the source IP and source MAC of the incoming frame.
 - As long as, a new IP (given that it is in the same subnet of the interface IP) has been detected, the arp table will form a new entry linking the MAC address in the same frame with this IP.

Quick Quiz



```
ip nat inside source static 192.168.1.1 200.1.1.5  
ip nat inside source static 192.168.1.3 200.1.1.7
```

```
ip nat inside source static 192.168.1.2 195.1.1.14  
ip nat inside source static 192.168.1.4 195.1.1.16
```

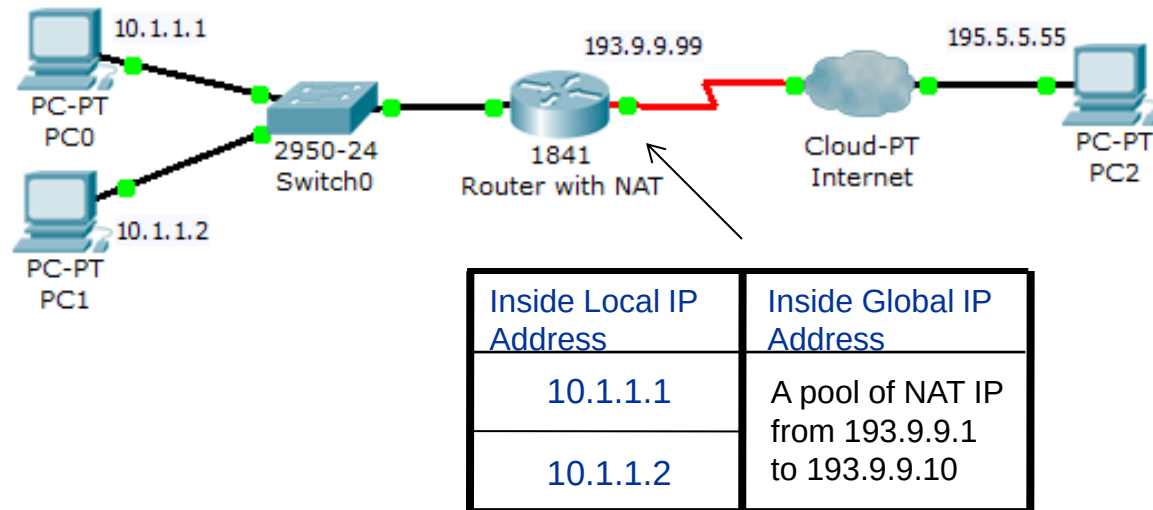
- The static NAT table of Router0 has been modified to translate to IP range of 200.1.1.x instead of 195.1.1.x.
- Can PC2 successfully ping 195.1.1.16?

Answer

- In this network: YES.
 - PC2 can ping 195.1.1.16
- Again, it is NOT recommended to translate static NAT source IP to any IP other than the IP range of the inside global.
- The problem is that the packet can reach the destination but may not be able to come back.
- However, it is a good “trick” to “masquerade” your source IP if you try to hack somebody.

Dynamic NAT

Working Principle of Dynamic NAT



- Dynamic NAT is designed to map a private IP address (local inside) to a pool of pre-defined public address (local global).
- Similar in some way compared to DHCP.

Configuring Dynamic NAT

- Step to configure dynamic NAT:
 - Create an access list (normally standard ACL)
 - Create a pool of NAT IP addresses
 - Link the access list (with hosts) to the pool.

```
Router(config)#access-list 10 permit host 10.1.1.1
Router(config)#access-list 10 permit host 10.1.1.2
Router(config)#access-list 10 permit host 10.1.1.3
Router(config)#ip nat pool DNAT 199.9.9.1 199.9.9.10 netmask 255.255.255.0
Router(config)#ip nat inside source list 10 pool DNAT
```

Access-list number

Last IP of the NAT pool

First IP of the NAT pool

Access-list number

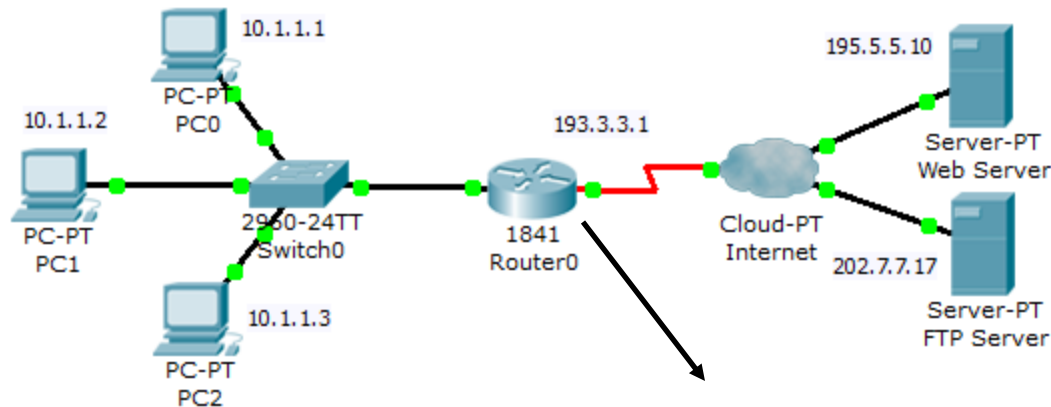
Name of Dynamic NAT pool

Comments on Dynamic NAT

- It is still a 1-to-1 IP change, but just draw from a pool instead of same “fixed” IP.
 - Which means the translated IP may be different each time for a particular PC. (e.g. sometimes you get 200.1.1.1, next time you will get 200.1.1.4).
- The ideal case is the number of hosts $\leq$ number of IP in the NAT pool.
- If the number of hosts $>$ number of IP in the pool
 - Not an efficient scheme, the host has to “wait” for the IP...
 - For example: 5 hosts with a pool of 3 IP. If the 3 hosts has “occupied” the NAT table, the other 2 PCs can’t travel to “outside network”.
- Design rule at the output interface of dynamic NAT is the same as static NAT.

Port Address Translation (PAT) or NAT overload

Working Principle of PAT



Inside Local IP Address	Inside Global IP Address	Outside Local IP Address	Outside Global IP Address
10.1.1.2:1456	193.3.3.1:1456	195.5.5.10:80	195.5.5.10:80
10.1.1.2:2199	193.3.3.1:2199	195.5.5.10:80	195.5.5.10:80
10.1.1.3:2333	193.3.3.1:2333	202.7.7.17:21	202.7.7.17:21

Sharing same global IP

- PAT uses unique source port numbers on the inside global IP address to distinguish between connections.

Features of PAT

- The most widely deployed form of NAT.
- Many inside local IP to one inside global (public) IP.
 - The only NAT method that helps to delay the IPv4 address exhaustion.
 - Static NAT and dynamic NAT DOES NOT help the IPv4 address exhaustion since both NAT methods still uses the same amount of global IP
- PAT is also known as NAT overloading
- PAT tracks the address translation in the form of IP:port instead of IP only (as in static NAT and dynamic NAT).

Configuring PAT (NAT overload)

- Steps to configure PAT (interface mode):
 - Use this when there is only one global IP
 - Create an access list
 - Put the access list on the “outside interface” with “overload”

```
Router(config)#access-list 11 permit 10.1.1.0 0.0.0.255
```

```
Router(config)#ip nat inside source list 11 interface s0/1/0 overload
```

- Steps to configure PAT (pool mode):
 - Use this when there are a few global IP
 - Create an access list
 - Create a NAT pool list as in dynamic NAT
 - Link the access list to the “NAT pool” with “overload”

```
Border(config)#access-list 12 permit 10.1.7.0 0.0.7.255
```

```
Border(config)#ip nat pool OUTBOUND 200.1.1.2 200.1.1.5 netmask  
255.255.255.0
```

```
Border(config)#ip nat inside source list 12 pool OUTBOUND overload
```

Limits of PAT

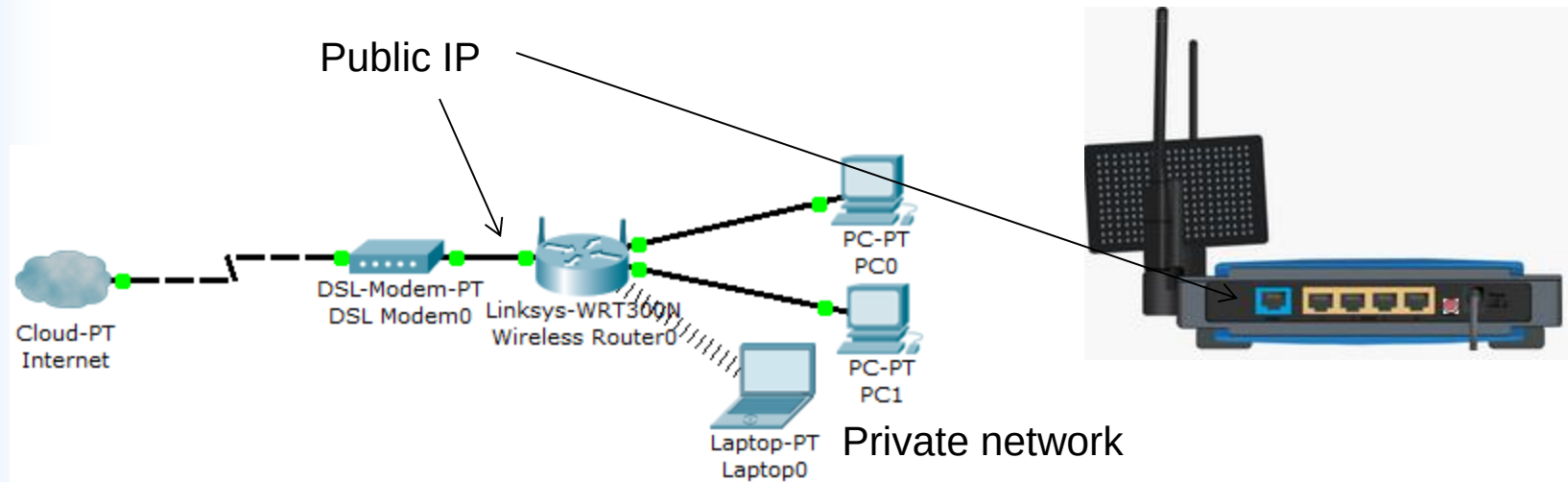
- Since PAT uses port number to differentiate between hosts of private networks, TCP and UDP port number range allows up to 65,536 number per global IP address.
- For PAT pool mode (you have a few global IP), if the ports for the 1st global IP is exhausted, PAT will look to the next IP address and try to allocate the original source port again.
- This will continue until all available ports and IP addresses are utilized.
- If a PAT router run out of “port number” and IP addresses, it drops the packet and sends an ICMP Host Unreachable message.

Checking PAT

- Use “show ip nat translations”
- The first two entries show either Dynamic NAT or Static NAT.
 - The entries with “---” on Pro, Outside local and Outside global.
- The last lines show PAT entries.

```
Border#show ip nat translations
Pro Inside global  Inside local  Outside local  Outside global
--- 200.1.1.25      10.1.1.25      ---            ---
--- 200.1.1.26      10.1.1.25      ---            ---
tcp 200.1.1.50:25    10.1.1.50:25    206.1.1.25:25  206.1.1.25:25
tcp 200.1.1.51:514  10.1.1.51:514   155.1.9.6:1021 155.1.9.6:1021
Border#
```

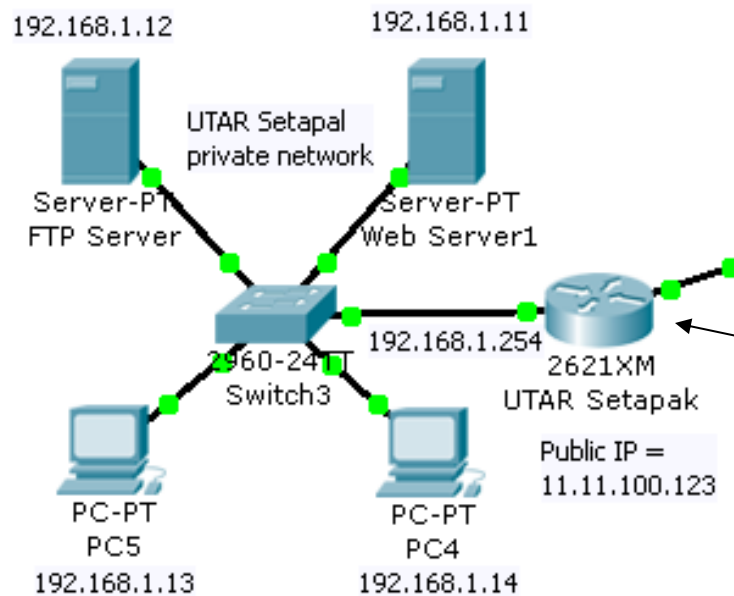

Wireless Router PAT



- The DSL wireless router at home does the PAT automatically for the user.
- Normally, the wireless router has a “internet port” in which the “Internet IP” is located.
- This “Internet IP” will be “shared” by both the wired network and the wireless network via the process of PAT.

Port Forwarding & Redirection

Private Network Server - 1



The world only see
The public IP

The world can't see
what is behind

- How do you let the world access the server in the private network?

Configuring Port Forwarding

- You can map static NAT in IP:port format as the port forwarding.
- You can map different local IP with different port to a global IP with different port.
- In the following example:
 - Packets with destination port 80 reaching interfaces with public IP will be redirected to 192.168.1.11
 - Packets with destination port 21 reaching interfaces with public IP will be redirected to 192.168.1.12

Inside network
Server IP

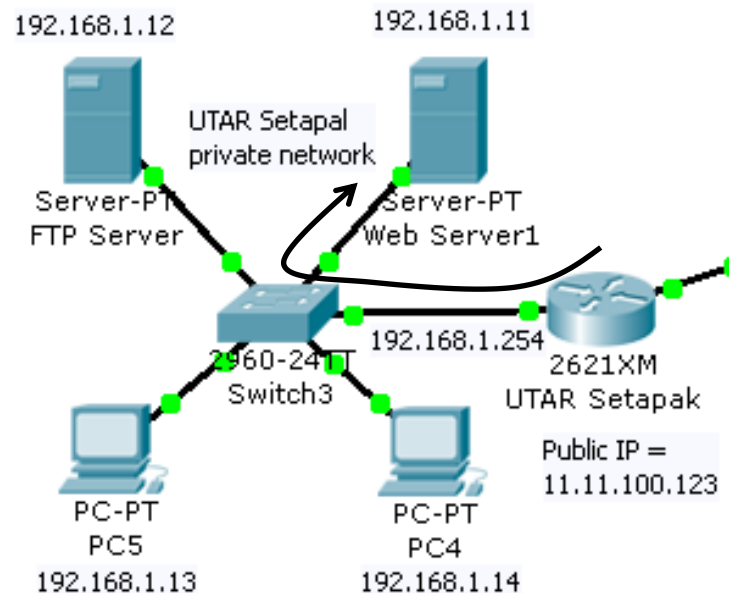
Public IP

```
R1 (config) #ip nat inside source static tcp 192.168.1.12 21 11.11.100.123 21
```

```
R1 (config) #ip nat inside source static tcp 192.168.1.11 80 11.11.100.123 80
```

Port number

Comments on Port Forwarding



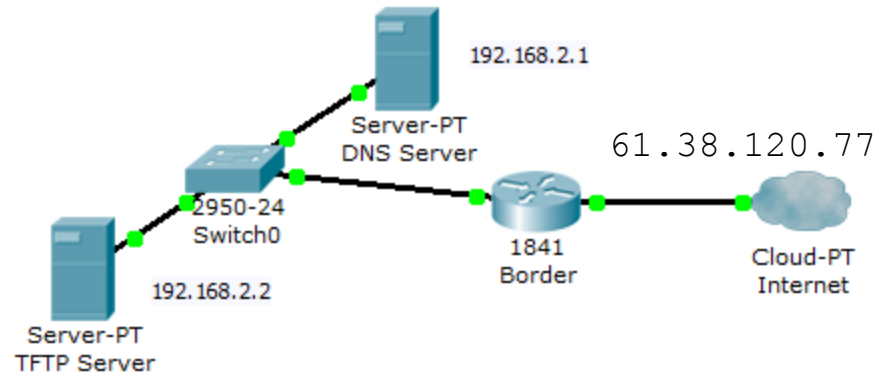
To the world, the web server:
<http://11.11.100.123>

However, the real web server
has an private IP of 192.168.1.11.

Packet received by the router
will be redirected to the web server.

- If the port forwarding has been set correctly, router will redirect the service to the inside server.
- Cons: Only 1 server for 1 port number.
- Note: As long as the router see the destination IP:port matches the tcp NAT entry, it will immediately convert the address and send out to appropriate local address.

UDP Port Forwarding

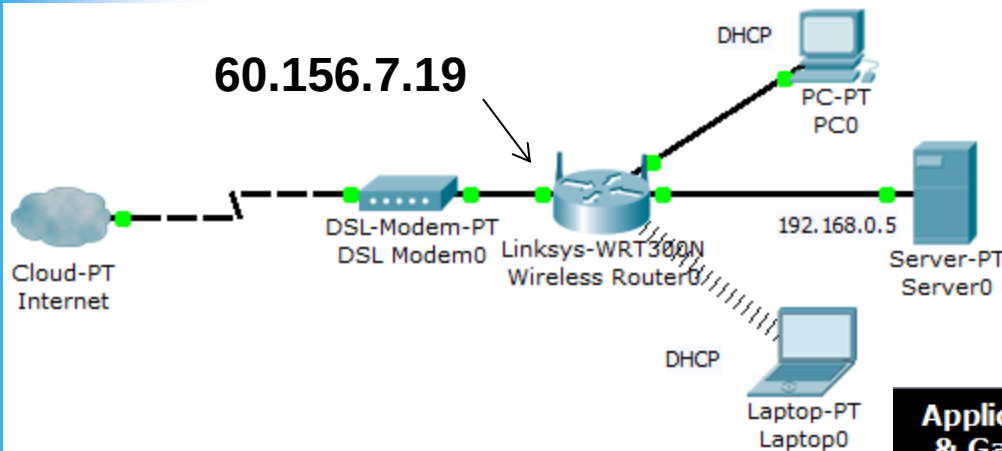


```
R1(config)#ip nat inside source static udp 192.168.2.1 53 61.38.120.77 53
```

```
R1(config)#ip nat inside source static udp 192.168.2.2 69 61.38.120.77 69
```

- Some services are done in UDP, hence the port forwarding is also set in UDP, especially service like:
 - DNS (port = 53)
 - TFTP (port = 69)

Configure Port Forwarding in Wireless Router



In this example, 3 services will be redirected to 192.168.0.5
 If the hosts of outside network do the following:
 c:>ftp 60.156.7.19
 c:>telnet 60.156.7.19
 http://60.156.7.19

- The configuration of port forwarding is shown as the right GUI screen with three services redirected to 192.168.0.5.

- HTTP
- Telnet
- FTP

Applications & Gaming

Setup Wireless Security Access Restrictions Applications & Gaming Administration

Single Port Forwarding Port Range Forwarding Port Range Triggering DMZ

Single Port Forwarding

Application Name

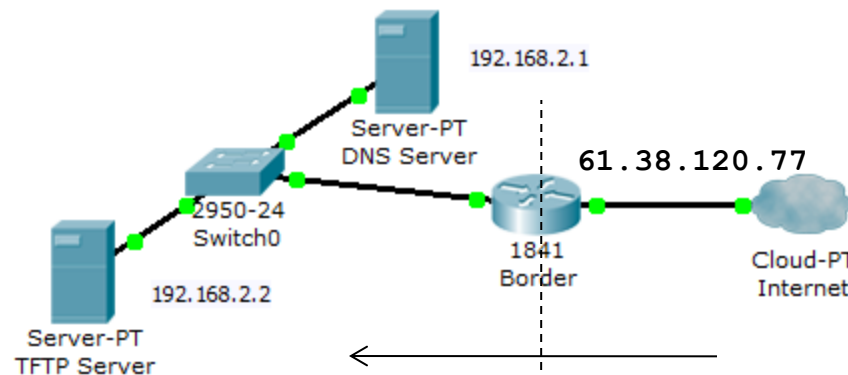
HTTP
Telnet
None
None
None

FTP

Externet Port	Internet Port	Protocol	To IP Address	Enabl
---	---	---	192.168.0.5	☒
---	---	---	192.168.0.5	☒
---	---	---	192.168.0.	☐
---	---	---	192.168.0.	☐
---	---	---	192.168.0.	☐
25	25	Both	192.168.0.5	☒

Comments on Port Forwarding

- Port forwarding is a special static NAT that translate the destination IP, when the packet travel from outside network to inside network via the NAT router.

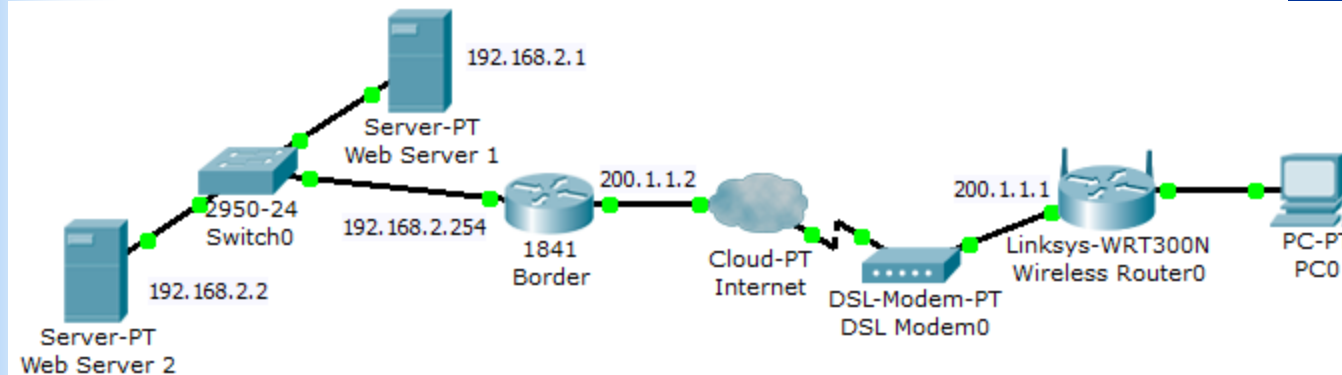


Source IP = 195.5.5.1
Destination IP = 192.168.2.1

Source IP = 195.5.5.1
Destination IP = 61.38.120.77

Extra configuration of NAT

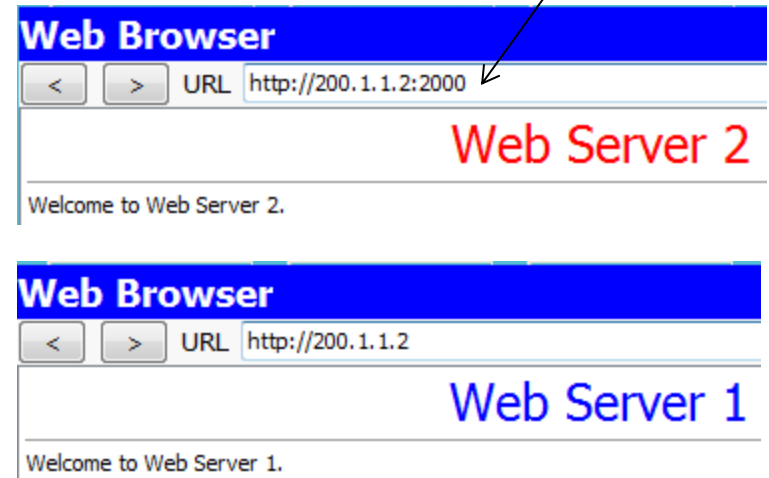
Adding Extra Private Web Server



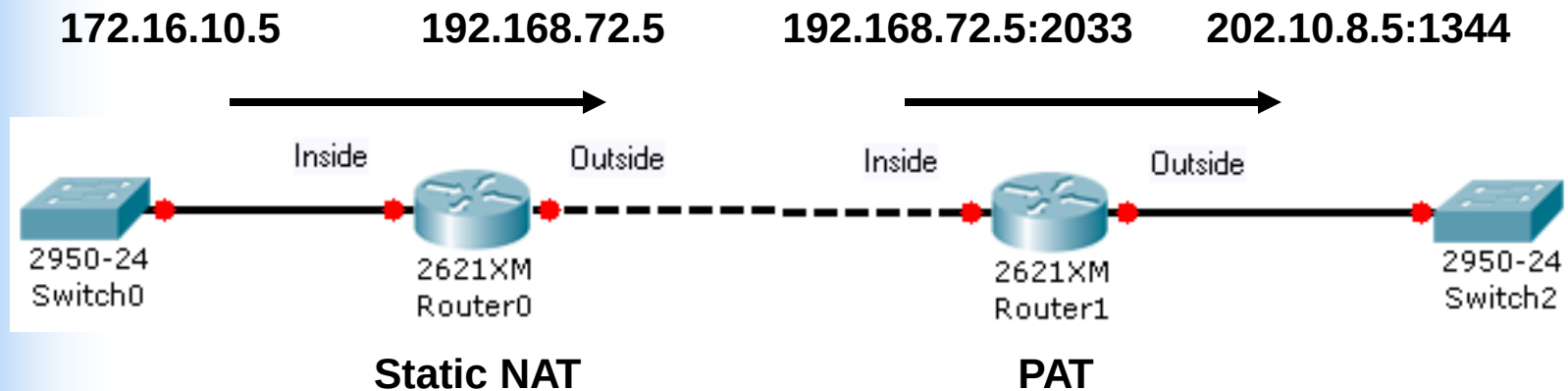
Pay attention to this port number 2000

```
R1(config)#ip nat inside source static tcp 192.168.2.1 80 200.1.1.2 80
R1(config)#ip nat inside source static tcp 192.168.2.2 80 200.1.1.2 2000
```

- Port forwarding only allows 1 port number to be redirected to 1 private IP.
- We can add extra private web servers by using a different external port number (e.g. 2000), so that the traffic can be re-directed to the extra web server, as shown in the example.
 - Accessing web server 2 (192.168.2.2) via `http://200.1.1.2:2000`

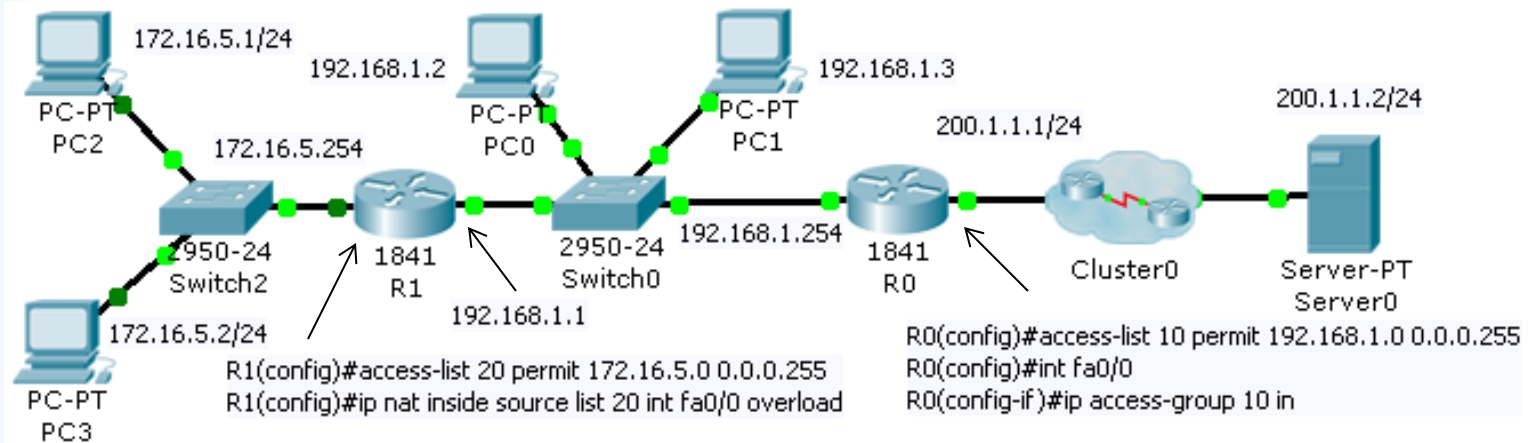


Double NAT / PAT



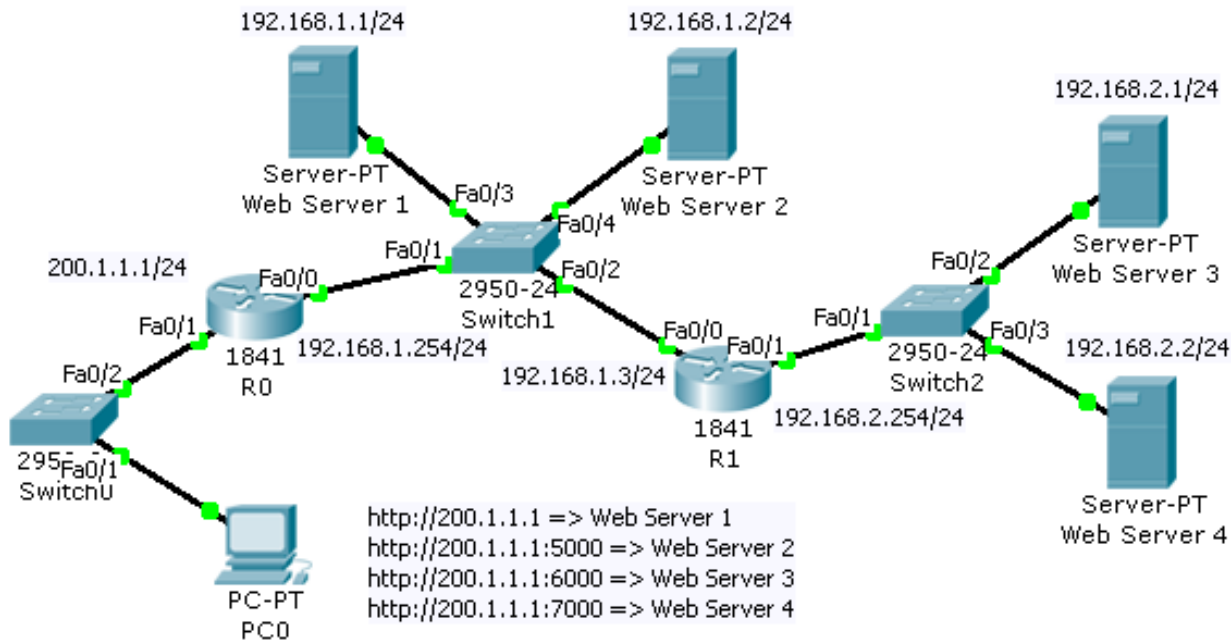
- Sometimes, we can design private network that can go through more than 1 stage of NAT (or PAT).
 - And the network still works properly.
- Is useful in some situation, especially for security purpose.

Example of Double NAT/PAT



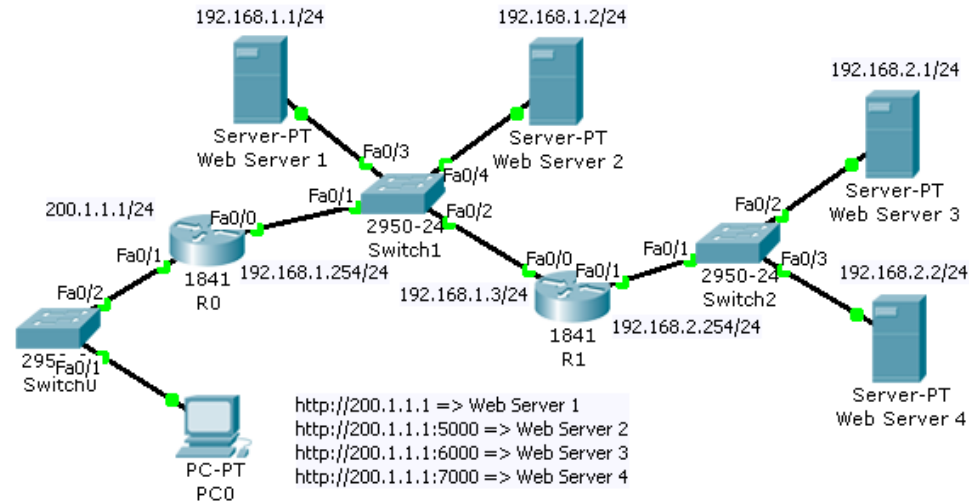
- R0 is the edge router, but only the traffic out from subnet 192.168.1.0/24 is allowed.
 - Assume that it is due to some security reason.
- Any subnet other than 192.168.1.0 has to undergo PAT.
- As in the example network, 172.16.5.0/24 has to masquerade as 192.168.1.1 with PAT, in order to pass through R0 to access Server0.

Double Port Forwarding



- How do you make PC0 to access all 4 private network servers via the public IP of 200.1.1.1?
- How do you configure Web Server 3 & Web Server 4 to be accessed from outside?

Double Port Forwarding



```
R0(config)#int Fa0/0
R0(config-if)#ip nat inside
R0(config-if)#int Fa0/1
R0(config-if)#ip nat outside
```

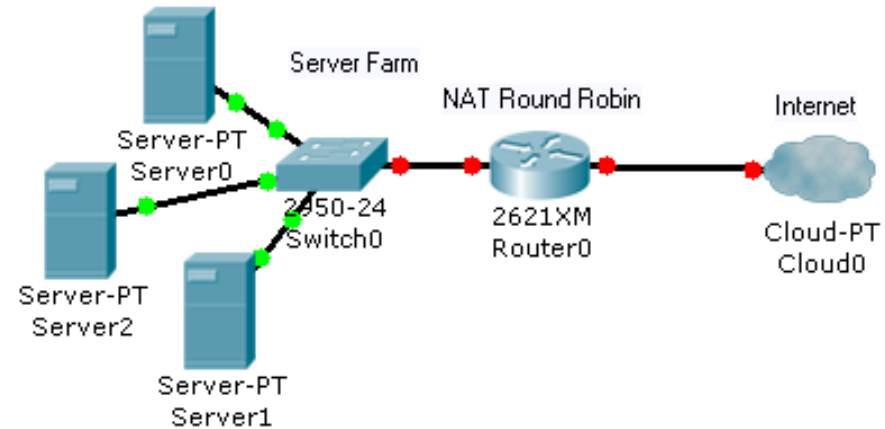
```
R0(config)#ip nat inside source static tcp 192.168.1.1 80 200.1.1.1 80
R0(config)#ip nat inside source static tcp 192.168.1.2 80 200.1.1.1 5000
R0(config)#ip nat inside source static tcp 192.168.1.3 6000 200.1.1.1 6000
R0(config)#ip nat inside source static tcp 192.168.1.3 7000 200.1.1.1 7000
```

```
R1(config)#int Fa0/0
R1(config-if)#ip nat outside
R1(config-if)#int Fa0/1
R1(config-if)#ip nat inside
```

```
R1(config)#ip nat inside source static tcp 192.168.2.1 80 192.168.1.3 6000
R1(config)#ip nat inside source static tcp 192.168.2.2 80 192.168.1.3 7000
```

Configuring for TCP Load Distribution

- Normally for server farms.
- Handle the load of incoming requests by spreading the load among several mirrored servers.
 - In round robin fashion
 - Not supported by Packet Tracer 5.3
- The following commands are example of NAT TCP load sharing



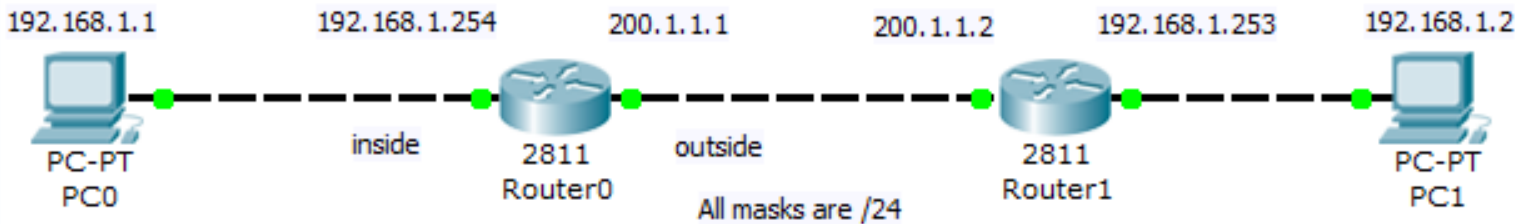
```
Border(config)#ip nat pool WEB-HOSTS 20.1.1.1 20.1.1.9 netmask 255.255.255.0
  ➡ type rotary
Border(config)#access-list 12 permit 20.1.1.254
Border(config)#ip nat inside destination list 12 pool WEB-HOSTS
```

- Note that the NAT pool is set with “type rotary”.
- It is “ip nat inside destination” instead of “ip nat inside source”.

IP nat outside source (1)

- **ip nat inside source.....**
 - Changing source IP from inside to outside,
 - Changing destination IP from outside to inside
 - Either inside to outside first or outside to inside first (will translate both side)
 - Most of the time, we use “ip nat inside source...”
- **ip nat outside source**
 - Changing source IP from outside to inside
 - Changing destination IP from inside to outside
 - Must be outside to inside 1st, then inside to outside
 - We hardly use “ip nat outside source....”

Quiz



```
Router0(config)#ip nat inside source static 192.168.1.1 200.1.1.1
Router0(config)#ip nat outside source static 192.168.1.2 200.1.1.4
```

```
Router0#show ip nat trans
```

Pro	Inside global	Inside local	Outside local	Outside global
---	200.1.1.1	192.168.1.1	---	---
---	---	---	200.1.1.4	192.168.1.2

Given the NAT lookup table of Router0, when PC1 ping 200.1.1.1, list the source IP and destination IP of the ICMP packet as it travel out and back.

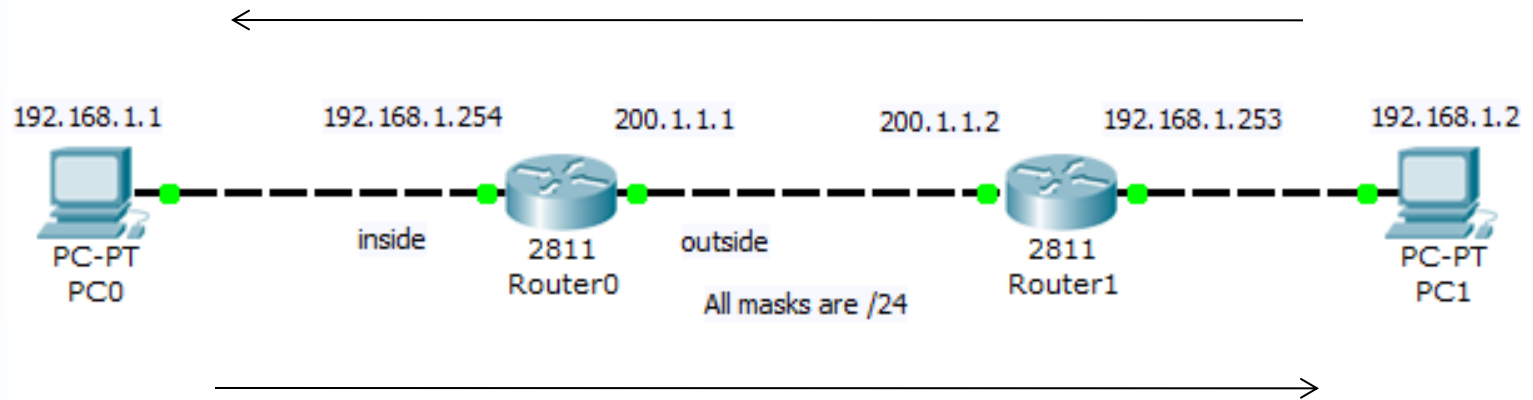
Router1 does not have NAT, and routes are properly set.

Answer

SIP = 200.1.1.4
DIP = 192.168.1.1

SIP = 192.168.1.2
DIP = 200.1.1.1

SIP = 192.168.1.2
DIP = 200.1.1.1

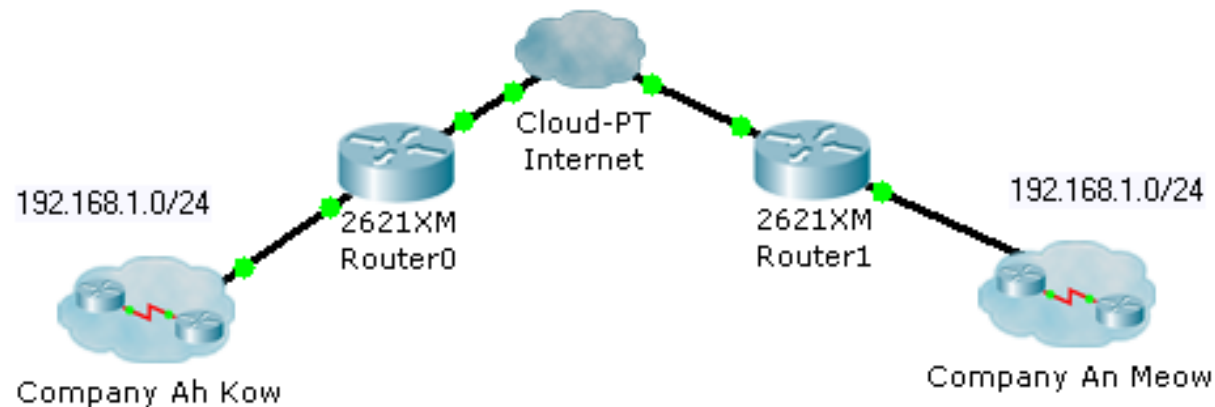


SIP = 192.168.1.1
DIP = 200.1.1.4

SIP = 200.1.1.1
DIP = 192.168.1.2

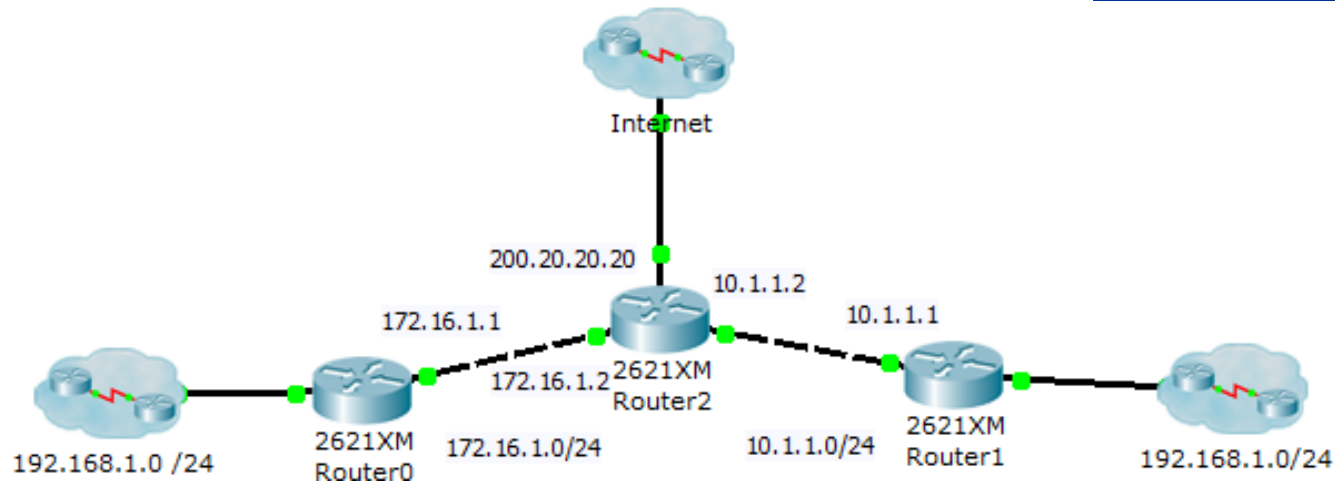
SIP = 200.1.1.1
DIP = 192.168.1.2

Scenario: Merging two Companies



- Two companies want to merge their network.
 - They want two network to form an intranet
 - They want to share 1 public IP to save cost
- However, their networks has similar network ID. Worse, they even have same IP addresses.
- It will be rather time consuming to convert all the IPs since their network has been running smoothly all these years.
- How would you solve this?

Scenario: Merging two Companies



- Set the NAT for both Router 0 and Router 1.
 - Set the left 192.168.1.0/24 to be translated to 172.16.1.0/24 using NAT.
 - Set the right 192.168.1.0/24 to be translated to 10.1.1.0/24 using NAT.
- Use PAT for Router 2.

PAT Default timeouts

- The router will keep PAT entries in the translation table for a configurable length of time.
- The router will remove translation table entries after certain amount of time.
 - Non-DNS UDP 5 min
 - DNS 1 min
 - TCP 24 hour
 - TCP RST/FIN 1 min

Troubleshooting NAT - 2

- Use the following commands to help you to troubleshoot problems in NAT.

Command	Meaning
<code>show ip nat translation</code>	Shows the basic translation table. This is probably one of the most important NAT command for verification.
<code>debug ip nat</code>	Shows the sending address, the translation, and the destination address on each debug line.
<code>show ip nat statistics</code>	Shows a summary of your configuration, your active translations, and the inside and outside interfaces that are being used.

Troubleshooting NAT - 2

- The “show ip nat translation verbose” command displays more information about each NAT table entry, such as the time left until the entry in the NAT table expires.

```
BorderRouter(config)#show ip nat translations verbose
Pro Inside global   Inside local Outside local   Outside global
--- 200.1.1.25      10.1.1.25      ---             ---
    create 00:05:01, use 00:00:00, left 23:12:40, flags: none
--- 200.1.1.26      10.1.1.26      ---             ---
    create 00:04:29, use 00:00:00, left 23:13:10, flags: none
```

Clear NAT Table

- Occasionally, NAT is properly configured but translations are not occurring.
- Most of the time, clearing the NAT translations resolves the problem.

Command	Meaning
<code>clear ip nat translation *</code>	Clears all NAT table entries.
<code>clear ip nat translation inside <i>global-ip</i></code>	Clears all inside NAT table simple translation entries.
<code>clear ip nat translation outside <i>local-ip</i></code>	Clears all outside NAT table simple translation entries.
<code>clear ip nat translation <i>protocol inside global-ip global-port local-ip local-port [outside local-ip local-port global-ip global-port]</i></code>	Clears all NAT table extended entries.

Quiz: NAT Table to Cisco Command

Pro	Inside global	Inside local	Outside local	Outside global
---	11.11.100.124	192.168.1.12	---	---
---	11.11.100.125	192.168.1.11	---	---
udp	11.11.100.123:1024	192.168.1.13:1025	11.11.11.11:53	11.11.11.11:53
udp	11.11.100.123:1026	192.168.1.13:1026	11.11.11.11:53	11.11.11.11:53
udp	11.11.100.123:1025	192.168.1.14:1025	11.11.11.11:53	11.11.11.11:53
tcp	11.11.100.123:80	192.168.1.11:80	---	---
tcp	11.11.100.123:23	192.168.1.12:23	---	---
udp	11.11.100.123:53	192.168.1.12:53	---	---
tcp	11.11.100.123:1025	192.168.1.13:1025	180.1.1.1:80	180.1.1.1:80
tcp	11.11.100.123:1026	192.168.1.13:1026	180.1.1.1:80	180.1.1.1:80

- Questions: Can you “convert” the NAT table back to Cisco command?

Answer:

```
access-list 20 permit host 192.168.1.11
access-list 20 permit host 192.168.1.12
ip nat inside source list 20 interface FastEthernet0/0 overload
ip nat inside source static tcp 192.168.1.12 23 11.11.100.123 23
ip nat inside source static tcp 192.168.1.11 80 11.11.100.123 80
ip nat inside source static udp 192.168.1.12 53 11.11.100.123 53
ip nat inside source static 192.168.1.12 11.11.100.124
ip nat inside source static 192.168.1.11 11.11.100.125
```

Quiz: Packets through NAT Table

```
Router#show ip nat translation
```

Pro	Inside global	Inside local	Outside local	Outside global
--	11.11.11.11	192.168.3.5	---	---
udp	11.11.11.11:53	192.168.3.6:53	---	---
tcp	11.11.11.11:1080	192.168.1.11:80	---	---
tcp	11.11.11.11:21	192.168.1.12:21	---	---
tcp	11.11.11.11:22	192.168.4.12:22	---	---

- A packet from outside network going to inside network;
 - Source socket = 11.12.12.12:1100, Destination socket = 11.11.11.11:1080;
- A packet from outside network going to inside network;
 - Source socket = 11.12.12.12:1100, Destination socket = 11.11.11.11:80;
- A packet from inside network going to outside network;
 - Source socket = 192.168.3.6:53, Destination socket = 11.12.12.12:1100;
- A packet from inside network going to outside network;
 - Source socket = 192.168.1.12:22, Destination socket = 11.12.12.12:1100;

Answer

- Source socket: 11.12.12.12:1100
- Destination socket: 192.168.1.11:80
- Source socket: 11.12.12.12:1100
- Destination socket: 192.168.3.5:80
- Source socket: 11.11.11.11:53
- Destination socket: 11.12.12.12:1100
- Source socket: 192.168.1.22
- Destination socket: 11.12.12.12:1100

NAT Benefits

- Eliminates re-assigning each host a new IP address when changing to a new ISP
 - Just reconfigure the NAT router
- Eliminates the need to re-address all hosts that require external access, saving time and money
- Conserves addresses through application port-level multiplexing
- Protects network security (some level of protection only)

Disadvantages of NAT

- NAT increases latency (delay)
 - Since it need extra processing
- NAT is resource intensive
 - Resource means CPU time and RAM
- NAT hides end-to-end IP addresses that render some applications unusable.
 - Local private IP can't communicate with remote private IP via Internet
- Because NAT changes the IP address, there is a loss of IP end-to-end traceability.
 - Makes troubleshooting or tracking down where malicious traffic is coming from more troublesome.

Some Supported Traffic of NAT

- TCP traffic that does not carry source and destination IP addresses inside the application stream
- UDP traffic that does not carry source and destination IP addresses inside the application stream
- Hypertext Transfer Protocol (HTTP)
- Trivial File Transfer Protocol (TFTP)
- File Transfer Protocol (FTP PORT and PASV command)
- Finger, a tool that determines whether a person has an account on a particular computer
- Network Time Protocol (NTP)
- Network File System (NFS)
- Many of the r* Unix utilities (rlogin, rsh, rcp)
- Internet Control Message Protocol (ICMP)
- NetBIOS over TCP (datagram, name, and session services)
- Progressive Networks' RealAudio
- White Pines' CuSeeMe
- Xing Technologies' StreamWorks
- DNS A and PTR queries

Some Unsupported Traffic of NAT

- Routing protocols
- DNS zone transfers
- BOOTP/DHCP
- Talk
- Ntalk
- Simple Network Management Protocol (SNMP)
- Netshow

Summary of NAT

- Enabling a private IP network to use unregistered IP addresses to access an outside network such as the Internet.
- Providing the ability to reuse assigned IP addresses that are already in use on the Internet.
- Providing Internet connectivity in networks where there are not enough individual Internet registered IP addresses
- Appropriately translating the addresses in two merged intranets such as two merged companies.
- Translating internal IP addresses assigned by old Internet service providers (ISPs) to a new ISP's newly assigned addresses without manually configuring the local network interfaces.

Concerns about NAT

- **Performance:**

- Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
- Modifying port number requires that NAT boxes recalculate TCP checksum

- **Fragmentation**

- Care must be taken that a datagram that is fragmented before it reaches the NAT device, is not assigned a different IP address or different port numbers for each of the fragments.

Concerns about NAT

- **End-to-end connectivity:**
 - NAT destroys universal end-to-end reachability of hosts on the Internet.
 - A host in the public Internet often cannot initiate communication to a host in a private network.

Concerns about NAT

- **IP address in application data:**
 - Applications that carry IP addresses in the payload of the application data generally do not work across a private-public network boundary.
 - Some NAT devices inspect the payload of widely used application layer protocols and, if an IP address is detected in the application-layer header or the application payload, translate the address according to the address translation table.

Last Comments on NAT

- History behind NAT:
 - It was first implemented in Cisco IOS release 11.2
 - A way to alleviate the depletion of the IPv4 address space.
 - NAT normally works together with private IP addresses.
- Future:
 - May be obsolete if IPv6 is used widely.